Supernova Neutrino Detection at ICNSE

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Neutrinos carry away most of the energy released in Supernova explosions and are messengers of stellar dynamics during the final evolutionary phases of massive stars

Motivation :

- **Understand the core collapse and explosion mechanism in supernovae**
- **Study of neutrino properties**

Introduction

A Supernova is powerful and luminous explosion of a star. It occurs in the final stages of a massive (>8 $M_{\odot}$) star, or when a white dwarf is reignited into runaway nuclear fusion. The explosion releases massive amounts of energy (>10^{50}erg), about 99% of which is carried away by neutrinos. These neutrinos contribute both to the explosion of the star as well as to cooling of the resulting proto-neutron star. Due to their weak interaction, neutrinos are clean messengers of core dynamics during the violent final stages of stellar evolution.

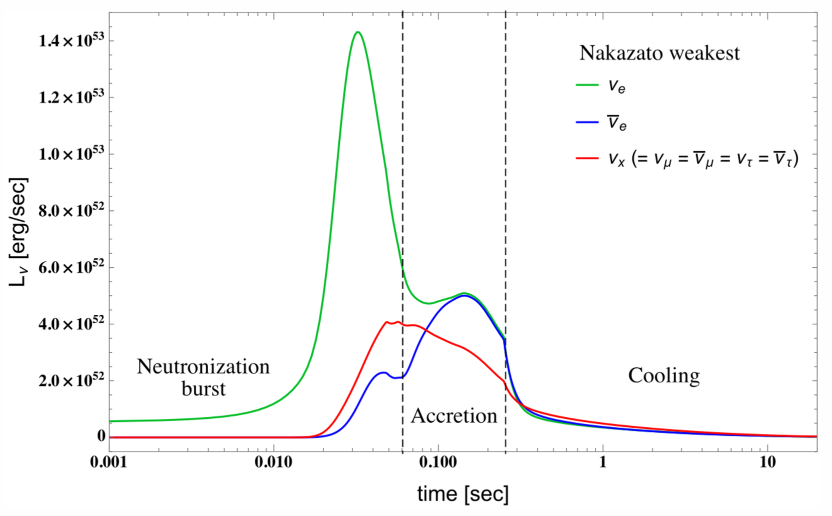


Fig 1: Time evolution of neutrino flux from a CCSN explosion of a 20 $M_{\odot}$ star

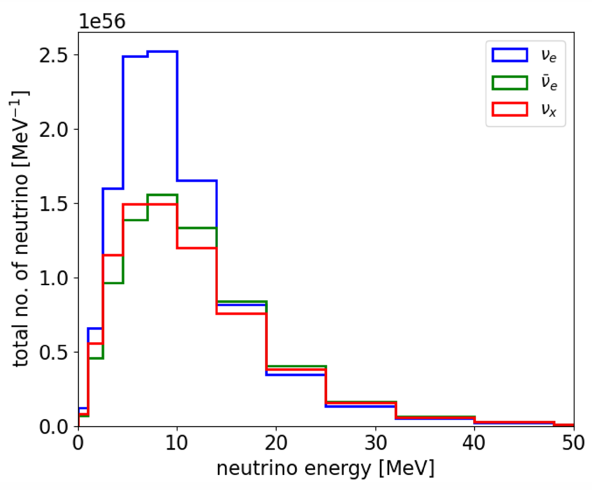
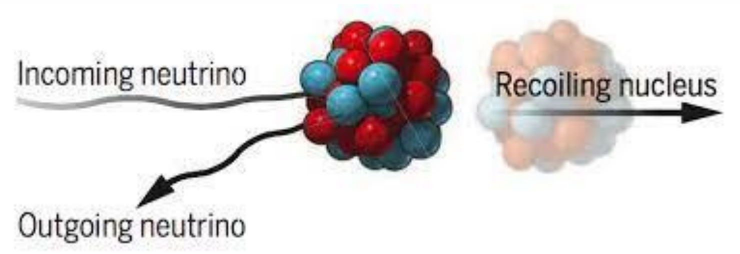


Fig 2: Time integrated of neutrino spectra from a CCSN explosion of a 30 $M_{\odot}$ star

The CEνNS Process



CEνNS is a

- Coherent: all nucleon Ψ's in phase
- Elastic: kinetic energy conserved
- Flavor-blind: same σ for all neutrino flavor

Scattering of ν off target Nuclei. Due to the coherent nature of scattering, the cross section is enhanced by ~N^2 of the target nuclei ($^{N+Z}_{Z}X$).

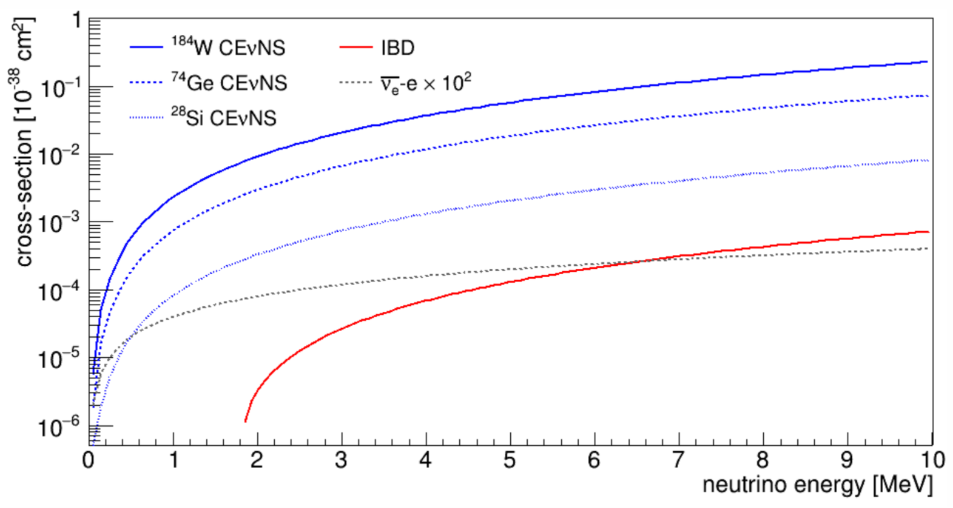


Fig 3: Comparison of CEνNS cross section for 3 nuclei with IBD and ν-e scattering.

ICNSE at APSARA-U

India based Coherent Neutrino Scattering Experiment (ICNSE) is proposed to be set-up at the APSARA-U experimental reactor at BARC, Mumbai, to study neutrino properties and BSM physics.

This detector will also be sensitive to supernova neutrinos (SNν).

The detector will consist of

- 30kg Ge/Si/Sapphire crystals
- operated at mK temperatures
- with phonon and ionization(for Si/Ge) or scintillation(for sapphire) readout
- housed inside multilayer shielding to reduce radiation background.

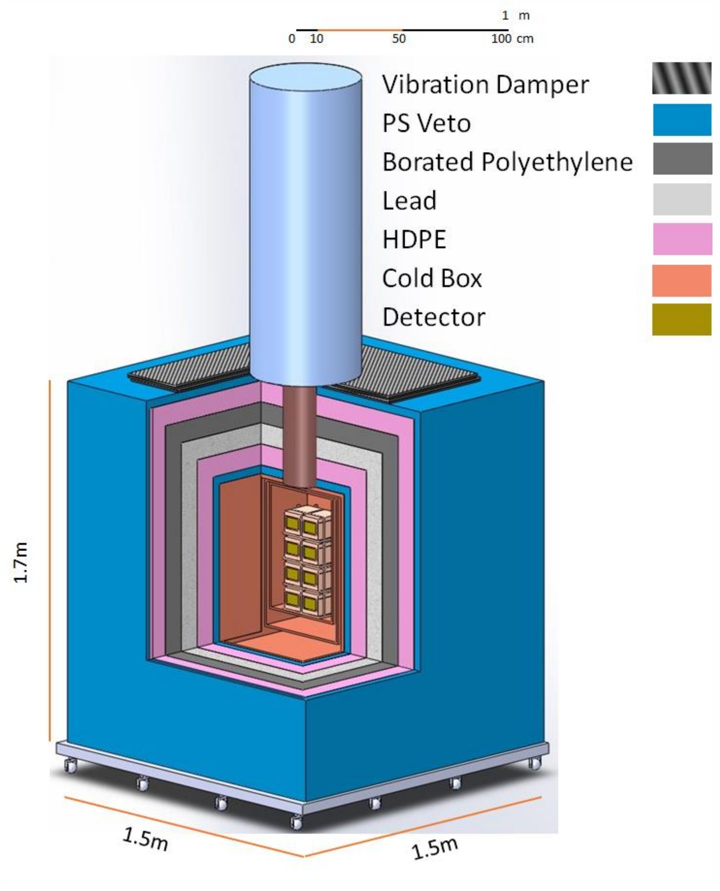


Fig 4: Concept design of ICNSE detector and shielding setup

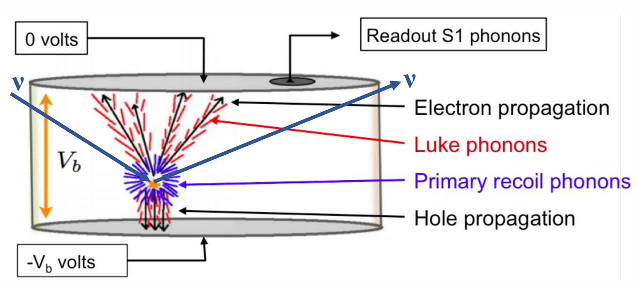


Fig 5: Detection principle of ICNSE detector

Results

- A code is written in python and used to evaluate the spectra of nuclear recoil energies in detector crystal using SNν spectra for different detector masses and supernova distances.
- Counts are taken above realistic threshold on recoil energy (in eVnr)
- For Ge, Lindhard quenching is also considered by normalizing the recoil energy to electron equivalent energy.
- For Lindhard quenching, threshold is set on electron equivalent energy (in eVee)

Detector (Threshold)	Counts/10kg/(196pc)^2
Sapphire (100eV)	229
Ge (10eVnr)	367
Ge (10eVee)	210

Table 1: expected SNν counts in ICNSE detector for different material and thresholds, for CCSN of 30 $M_{\odot}$ star.

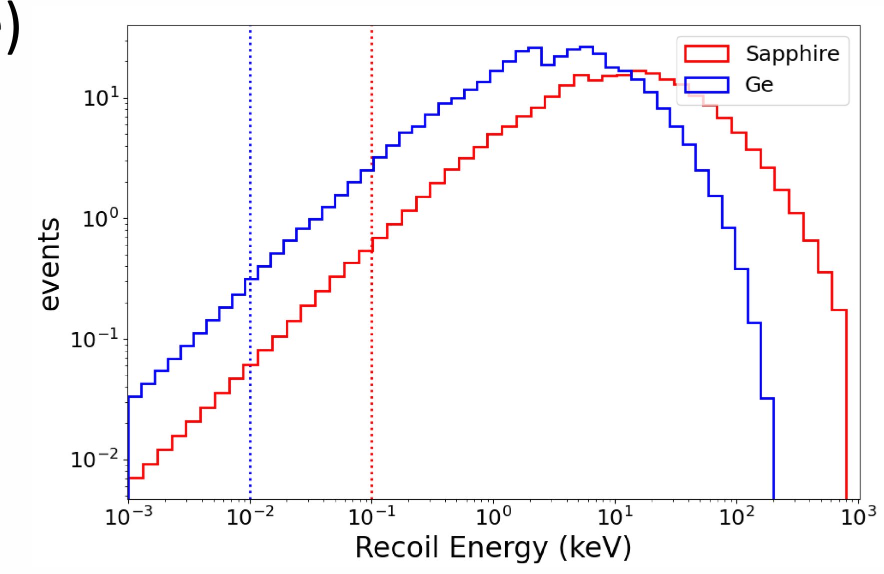


Fig 6: Nuclear recoil spectrum from CCSN of 30 $M_{\odot}$ star at 196pc in 10kg Ge and Sapphire

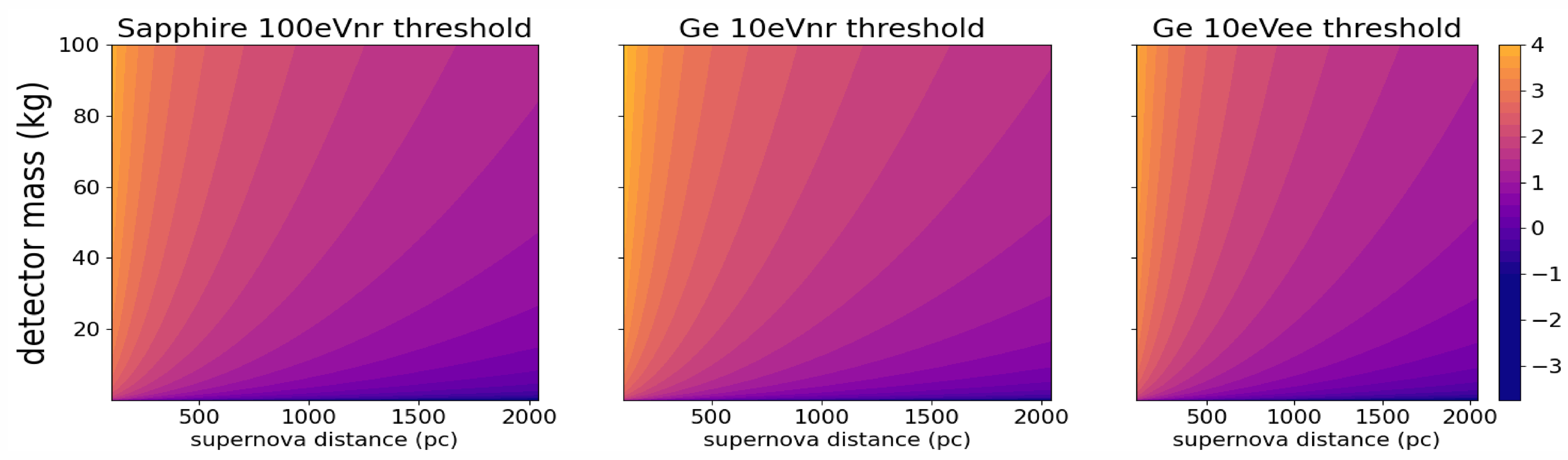


Fig 7: Contour plot of $\log_{10}$(SNν count) from CCSN of 30 $M_{\odot}$ star at different distances (x-axis) and detector mass (y-axis) for Sapphire (left) and Ge. The center plot shows counts for threshold on recoil energies, while right plot is with lindhard quenching.

Future Outlook:

- Sensitivity study for different models for supernova explosion